The Bruhat–Tits Tree Code — Status and Test Plan

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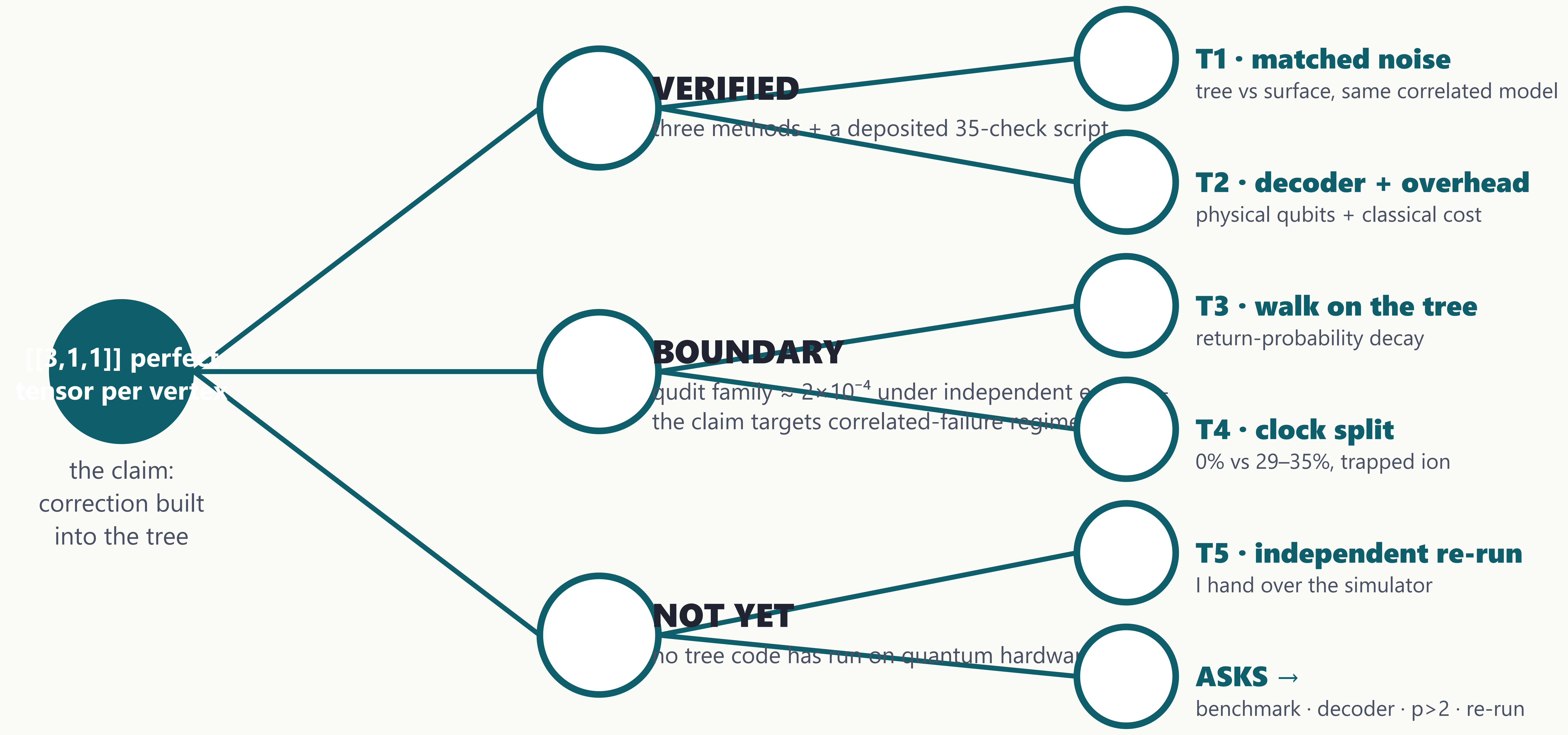

The tree is the code: the claim at the root, where it stands on the three branches, the five tests that would change my mind on the leaves.

50.0%
bit-flip
4.6× surface (~10.9%)

75.0%
depolarizing
75× surface (~1.0%)

17.30%
independent X+Z
no direct surface figure

The tests

T1 · Matched-noise comparison
Tree vs surface under the *same* correlated noise model. Classical, days–weeks.
Would change my mind: the advantage falls to parity.

T2 · Decoder + full overhead
Physical qubits plus classical decoding cost. Classical.
Would change my mind: the advantage disappears once decoding cost is counted.

T3 · Random walk on a p-regular tree
Return-probability decay. Ion / analog simulator, ~8 weeks.
Would change my mind: no change in the decay exponent where predicted.

T4 · Clock-rest split
Diagonal 0% vs non-diagonal 29–35% violation rate. Trapped ion, ~8 weeks.
Would change my mind: the measured split falls outside that range.

T5 · Independent re-run
An outside group repeats T1 with my simulator and the protocol.
Would change my mind: nothing to predict — this one is about reproducibility.

For this room

1. A correlated-noise benchmark — model and data — I can run T1 against.
2. A decoder reality check: does a known decoder family already cover tree-structured codes, and how does it scale?
3. Pointers on perfect tensors for $p > 2$ (the binary [[3,1,1]] case is settled).
4. An independent re-run of T1.

Verified: thresholds computed three independent ways — analytical recursion, exact stabilizer simulation, Monte Carlo — and re-checked by a deposited 35-check script.

Boundary: under independent errors the qudit family sits at $\approx 2\times10^{-4}$, about 55× below surface codes; the advantage claim targets correlated-failure regimes — which is exactly what T1 probes.

All numbers are classical simulation — i.i.d. noise, crossing criterion; nothing has run on hardware yet. · DOIs (verified 21 Aug 2026): 10.5281/zenodo.20109836 (architecture) · 10.5281/zenodo.22038733 (guide + verification script) · 10.5281/zenodo.21046993 (qudit QEC) · 10.5281/zenodo.22025544 (trapped-ion testbed) · 10.5281/zenodo.21964674 (QEC–Darwinism audit) · 10.5281/zenodo.21979060 (prime-valuation correction) · Full plan: github.com/QNFO/cwi-qec-poster-2026 → docs/tree-code-roadmap.md